
BOARD DIPLOMA EXAMINATION, (C-23)

DCME - FIRST YEAR EXAMINATION

PROGRAMMING IN C (C23-CM-106)

Time: 3 Hours

[Total Marks: 80]

PART-A

List the basic data types in C with one example each. (3)

Define a constant and a variable in C. (3)

Write the syntax for formatted input using scanf(). (3)

List three logical operators in C with their symbols. (3)

What is a switch statement? Write its syntax. (3)

Define a nested loop and give one example. (3)

Explain how to declare a two-dimensional array in C. (3)

Define a structure in C and give one example. (3)

What is a function prototype? Give an example. (3)

Define a pointer and write the syntax to declare a pointer variable. (3)

PART-B

Explain the steps involved in editing, compiling, executing, and debugging a C program. (10)

Explain the precedence and associativity of operators in C with examples. (10)

Write a C program to find the sum of digits of a given number using a while loop. (10)

(a) Write a C program to reverse a string using string handling functions. (6)

(b) Explain the difference between break and continue statements with examples. (4)

Write a C program to perform matrix addition using two-dimensional arrays. (10)

Explain the concept of structures and unions, highlighting their differences with examples. (10)

Write a C program to demonstrate a function with arguments and return value to calculate the factorial of a number. (10)

Explain file handling in C, including file opening, closing, and input/output operations with a sample program. (10)